VIDYUT

AI-Native Grid Intelligence for India's Renewable Transition

Predicting where renewable power will be wasted. Locating exactly where it's being stolen or lost.

Founders

Aashna Suman • Madhav Tiwari

AI Founders Lab — Pre-Seed Concept Pitch

THE PROBLEM

India's grid is leaking money in two directions at once

Renewable Curtailment

2.1 TWh

of clean electricity curtailed in FY 2025-26 — about ₹629 crore in foregone revenue — to keep coal plants running at minimum load.

Grid Losses & Theft

22.5%

technical losses on distribution networks — well above the 5-10% global benchmark — plus tens of billions of dollars lost annually to electricity theft.

The common root cause: existing tools monitor each substation or meter in isolation. But curtailment and theft are network phenomena — they show up as patterns across connected points, not at one point alone.

THE SOLUTION

One pipeline, two AI products

A clamp-on sensor streams live grid data. Graph-based AI reasons over how the network actually connects. A live dashboard delivers the answer before the loss happens.

1. Sense

Clamp-on hardware sensor streams live voltage, current & weather data continuously.

2. Predict

Graph-based AI models reason over the grid's real topology to forecast and localize.

3. Deliver

Live dashboard + field app show operators exactly what's about to go wrong, and where.

Framing note: *this is two AI products, not three offerings. Hardware and dashboard are the input/output layers of each product, not standalone products.*

PRODUCT 1

Curtailment Prediction

Forecasts where and when a renewable bottleneck will occur — ahead of time — so operators can reroute, store, or trade power instead of switching it off.

1

1. Sense

HARDWARE · NO AI

Clamp-on sensor + micro-weather inputs stream live grid and generation data continuously. This is the live-monitoring foundation.

2

2. Predict

CORE IP · AI

A temporal Graph Neural Network models the grid as a graph — substations as nodes, lines as capacity-weighted edges — learning how load and generation propagate to forecast where and when curtailment will hit.

3

3. Deliver

SOFTWARE · NO AI

Live dashboard shows real-time readings plus forecasted bottlenecks and alerts, with a track record of predicted-vs-actual outcomes.

PRODUCT 2

Fault & Theft Localization

Infers the physical origin point of a fault or theft from correlated anomaly signals — not just flagging that something looks wrong.

1

1. Sense

HARDWARE · NO AI

Clamp-on CT sensor at transformer/meter clusters streams live voltage & current data — the same live-monitoring foundation, reused across the low-voltage network.

2

2. Predict

CORE IP · AI

An anomaly detection model flags unusual consumption/voltage patterns; a graph-based localization model then triangulates the true origin point from multiple correlated signals, using real network topology.

3

3. Deliver

SOFTWARE + FEEDBACK

Dashboard flags the location on the network map; technicians confirm/reject via mobile app, and that feedback retrains both models over time.

WHY IT'S DEFENSIBLE

Two hard ML problems. Nothing else.

Only the middle stage — Predict — is genuinely hard, defensible AI work. Everything else feeds it better data or delivers its output.

Graph-native, not per-node

The grid is modeled as an actual graph with real topology — a bottleneck or anomaly is treated as a network event, not an isolated point statistic.

Prediction, not detection

Forecasts where and when a problem will occur, giving operators time to act — instead of a dashboard that lights up after the fact.

Localization, not just flagging

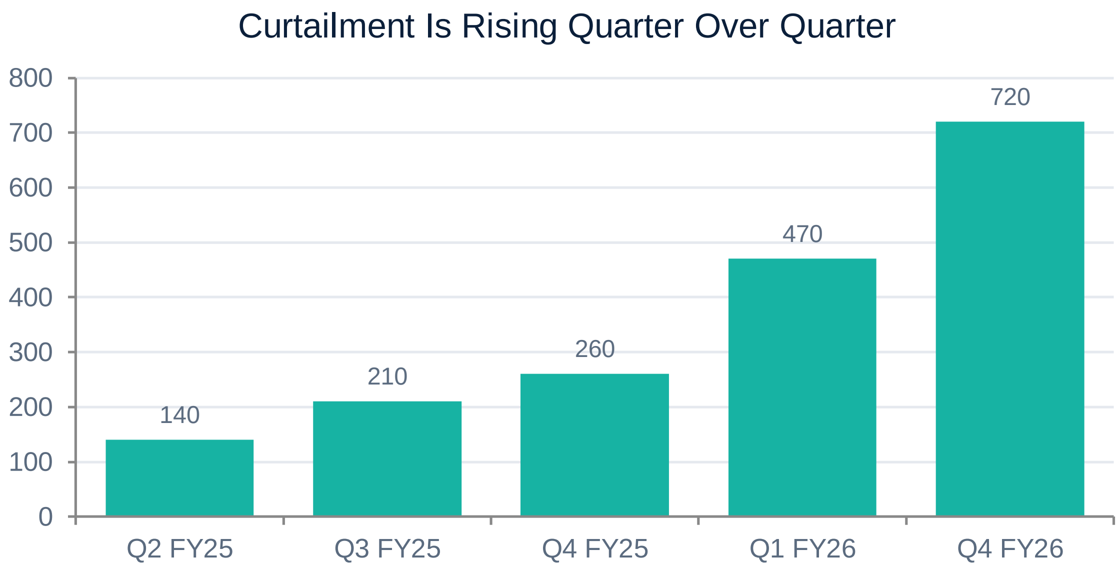
Pinpoints the physical origin of an anomaly using topology — the harder, far less commoditized problem versus simple flagging.

Compounding accuracy

Field-confirmed feedback retrains both models continuously — accuracy improves with deployment instead of staying static.

MARKET OPPORTUNITY

A large and growing problem, on both sides



Illustrative trend based on Ember/CREA quarterly reporting (FY25-FY26); see appendix for sourcing.

50 GW renewable capacity added in India in 2025 alone

500 GW non-fossil capacity target by 2030 — ~₹30.5 lakh crore investment needed

60+ state & private discoms under pressure to cut AT&C losses

25 crore smart meters targeted under the national rollout — a live data foundation to build on

How Vidyut earns, and how it lands

REVENUE MODEL

- **SaaS / B2G licensing**

  Per substation-node or per-meter-cluster monitored, sold on annual contracts to discoms and grid operators.

- **Hardware bundle**

  Sensor pods and edge appliances sold or leased alongside software for sites without existing telemetry.

- **Value-based pricing**

  Priced partly against quantifiable savings — avoided curtailment payouts, recovered theft revenue.

GO-TO-MARKET

- **Pilot on one feeder**

  One mid-size discom or SLDC, using existing SCADA/smart-meter data — no new hardware required to prove the model.

- **Prove it with numbers**

  Use quantified pilot savings (MWh avoided, ₹ recovered) as the case study for the next 3-5 sales conversations.

- **Layer in hardware selectively**

  Add sensor pods only where existing telemetry is too sparse for the models to perform well.

Everyone solves half the problem

SCADA / AMI Vendors

Grid telemetry & smart-meter infra

Deliver raw monitoring and connectivity — no predictive layer on top of the data.

Anomaly Analytics Vendors

Meter-level flagging tools

Flag unusual readings, but stop at detection — no localization, no network reasoning.

Manual Vigilance

Discom raids & field teams

Physical raids and static heuristics — reactive, labor-intensive, no forecasting.

Vidyut's gap: no existing player combines live sensing with graph-based prediction of where a bottleneck or anomaly will occur. Our closest competitive risk is a well-funded AMI vendor or global grid-analytics platform expanding into this space.

TEAM

Built by students, not around one

Two B.Tech founders, close enough to the technical layer to build the core models and product ourselves — and young enough to commit fully to one problem.

AS

Aashna Suman

AI / ML & Research

- Published anomaly-detection framework (IMMUNE) — the same core technique behind Product 2's detection layer
- Prior research spanning graph-based systems and RL-based decision models in the energy domain
- Four published research papers; concurrent AI specialization alongside CS degree

MT

Madhav Tiwari

Product Engineering & Design

- iOS and web development — builds the live dashboard and field-technician app in-house from day one
- Photography and visual design background — brings genuine product polish, rare in grid-tech tooling
- Covers the full delivery layer without depending on outside technical hires

Next Steps

From concept to a live pilot

1. Scope the training data pipeline for both core models — confirmed data vs. what must be bootstrapped in a pilot
2. Identify 2-3 target discoms or SLDCs for pilot conversations and confirm data-sharing feasibility
3. Pressure-test the defensibility claim against published literature so it survives technical due diligence

Vidyut

Aashna Suman • Madhav Tiwari